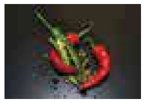
Restoring vision via gene therapy
Using gene therapy, scientists have been able to restore vision in blind mice. **http://dgit.in/nov20-23**



Device to rate pepper's hotness
Scientists have developed a portable, sensor device to detect capsaicin levels in pepper samples. **http://dgit.in/nov20-24**

HOW DO PLANES FLY?

A propellor plane flying upside down

Credit: Knumina Studios / Shutterstock.com

Scientists have not yet found a complete explanation for what generates lift, and fight over which of the partial explanations are true.

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When a plane is in flight, there are four basic forces acting on it that have to do with keeping it in the air. While moving and at rest, there is the constant force of gravity between the plane and ground. When an object moves through a fluid medium, the medium resists the motion, and this resistance is a force known as drag. For the purposes of understanding drag, it is important to treat air as a fluid. The aerodynamic shapes of racing cars, fighter jets and speed boats all have to do with reducing the amount of drag on an object. If for example you stick your finger out of a car window, you will not experience much of a force against it, but if you stick out your arm, the air exerts a considerable force against the hand, pushing it backwards. This is drag, and it is beneficial to reduce drag as much as possible to achieve speed. The poses adopted by swimmers and skiers allows them to reduce the drag as they move.

Then there is the thrust, which moves the plane forwards through the air. This is provided by the propellers or the jet engines on the plane. The thrust occurs in the opposite direction as the drag, and in fact it is the thrust that causes the drag.

Finally, there is the most important force of all, which is called the lift. The wing of a plane is technically referred to as an aerofoil, and it is this that produces the lift. Lift is a force that pushes the plane upward, and is a phenomenon associated with the movement of air around the aerofoil. The largest plane in the world, the Antonov An-225 Mriya has a take off weight of 640,000 kg. The wings have to produce enough lift to keep this monster in the air. The problem with understanding how aeroplanes fly, is in finding a complete explanation for what produces lift. The best theories we have at present are unable to account for a difference in air pressure at the top and bottom of the aerofoil. There is a zone of low pressure created on the top surface, that pulls the wings upwards, and while scientists know that it is formed when the plane is in flight, they are unable to explain it completely.

There are a number of theories that explain lift, which are so oversimplified that they are straight up wrong. One is known as the Longer Path or Equal Transit theory. According to this theory, the curved shape of the